Leakage-Controlled Zero- and Few-Shot Drug Repurposing for Rare Autoimmune Diseases: A Transfer Study with a Path-Based Graph Neural Network

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Abstract—Rare autoimmune diseases have few approved drugs, so computational repurposing must rank drugs for diseases whose treatments are unknown. We study this setting on the PrimeKG knowledge graph under a leakage-controlled protocol that holds out each test disease together with its related diseases and ranks all 7,957 drugs. Our model, BioPathNet++, reuses the path-based BioPathNet architecture and adds a channel of four leakage-free priors; we do not propose a new layer. The contributions are the protocol, a controlled transfer study, a few-shot evaluation, a head-to-head with TxGNN and external validation. On 68 unseen test diseases a known indication is in the top 10 for 76% of diseases (random: 0.8%), and the mean reciprocal rank (MRR) of 0.501 exceeds every baseline, including knowledge-graph embeddings (best baseline 0.392). When training data is scarce, autoimmune source diseases transfer better than label-matched controls (0.290 vs. 0.126); at full scale they even lower MRR (0.410 vs. 0.532). Revealing three known drugs raises MRR from 0.313 to 0.533 without retraining, and fusing the model with co-indication filtering gives 0.607 (best baseline 0.574). On TxGNN's own split BioPathNet++ ranks all drugs significantly better. Its outputs are hypotheses for expert review.

Index Terms—Drug repurposing, PrimeKG, knowledge graphs, graph neural networks, zero-shot learning, few-shot learning, transfer learning, rare autoimmune diseases, data leakage

I. Introduction

DRUG repurposing identifies new indications for approved drugs. Because these drugs already have known safety profiles, development is shorter and cheaper than for new molecules [1]. Rare diseases need repurposing most: of more than 7,000 rare diseases only 5–7% have an FDA-approved drug [2], and together they affect 3.5–5.9% of the population [3]. Autoimmune disorders are a related unmet need: 19 common autoimmune diseases affected 10.2% of a 22-million-person UK cohort [4], and many individual autoimmune conditions are rare and have few disease-specific drugs. We study six of them: primary biliary cholangitis (PBC), systemic sclerosis (SSc), Sjögren syndrome, autoimmune hepatitis (AIH), myasthenia gravis (MG) and Crohn disease.

Biomedical knowledge graphs (KGs) integrate drugs, proteins, phenotypes, pathways and diseases, and link prediction on them proposes drug–disease pairs [5]–[8]. Most work evaluates on randomly held-out edges, which rewards copying the drugs of near-identical diseases. The clinical question is different: what could treat a disease for which nothing, or almost nothing, is known to work? TxGNN [2] formalised this zero-shot setting by holding out whole diseases, and BioPathNet [9] showed that path-based reasoning with Neural Bellman–Ford networks (NBFNet) [10] ranks zero-shot indications well on PrimeKG [11].

We use transfer in this zero-shot sense: knowledge learned from diseases with known treatments is applied, without retraining, to diseases whose treatments the model has never seen. Autoimmune diseases share immune pathways, so the most useful source of that knowledge for rare autoimmune diseases might be the data-rich ones: rheumatoid arthritis (RA), systemic lupus erythematosus (SLE) and multiple sclerosis (MS). We ask three questions. (1) Source of transfer: must the knowledge come from related autoimmune diseases, or does broad pharmacology suffice (Section VIII-B)? (2) Accuracy: does a path-based model beat simple repurposing rules and KG embeddings on unseen diseases (Section VIII-A)? (3) Components: what do path reasoning, the prior channel, the loss and the graph contribute (Section VIII-D)? Because a few drugs are often known even for a rare disease, we also ask how much a handful of known drugs helps (few-shot, Section VIII-C).

A. What is new relative to BioPathNet

BioPathNet++ is not a new architecture. Its path channel is BioPathNet's NBFNet design [9], [10], and the name only marks the additions in Table I. The novelty of this work lies in the questions it answers and in how it measures the answers:

- a strict-family zero-shot protocol that also hides each test disease's related diseases, motivated by a leakage audit of our own earlier pipeline, in which a degree heuristic matched a trained GNN (Section IV);
- a controlled transfer study that compares autoimmune source diseases with label-matched non-autoimmune controls, from 11 training diseases to the full graph;

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- a few-shot evaluation that reveals one to three known drugs of an unseen disease, with collaborative-filtering baselines;
- a prior channel of four leakage-free features and a memory-efficient full-graph implementation that trains on one 16 GB GPU;
- a head-to-head with TxGNN on TxGNN's own split, which shows that the winner depends on the set of drugs being ranked;
- stronger baselines (KG embeddings, disease kNN, collaborative filtering), repeated seeds, and external validation of new candidates against trials, PubMed and later drug approvals.

TABLE I
What BioPathNet++ reuses from BioPathNet and what this work adds ("–": not part of the cited comparison).

Aspect	BioPathNet [9]	This work
Path model	NBFNet on a biomedical KG	reused (6 layers, DistMult messages, PNA scaling)
Extra signal	–	prior channel: 4 leakage-free features
Objective	negative sampling	softmax over all 7,957 drugs
Zero-shot unit	TxGNN disease areas	disease + its related diseases; per-batch hiding
Scale	background regulatory graph	full 4M-edge graph on one 16 GB GPU
Transfer source	not studied	matched controls, 11 diseases to full graph
Few-shot	–	1–3 revealed drugs, CF baselines
TxGNN comparison	on TxGNN's splits	TxGNN's code re-run, 3 seeds, two candidate sets

II. Related Work

Knowledge graphs for repurposing. Hetionet integrated 29 public resources and used metapath features to prioritise candidates [5]. DRKG [12] and PrimeKG [11] followed; PrimeKG covers 17,080 diseases with indication, contraindication and off-label edges. Recent reviews trace the field from classical machine learning to graph neural networks [7] and catalogue multi-relation learning methods [8]; Bonner et al. show that KG-embedding results depend strongly on dataset construction and evaluation [6], [13].

Embedding and GNN methods. Translational and bilinear embeddings (TransE, DistMult, RotatE) [14]–[16] and relational GCNs [17] are standard link predictors. They were applied to COVID-19 [18], [19], polypharmacy side effects [20], KG-Predict [21] and DREAMwalk, which reports up to 16.8% higher drug–disease association accuracy than link-prediction models [22]. A unified benchmark showed that many reported embedding gains shrink under consistent training and evaluation [23]. Embeddings learn one vector per node from its training edges, so a disease without known drugs gets little supervision. KGiA augments KGs with disease-specific counterfactual edges to reach previously unseen entities [24].

Zero-shot and path-based methods. TxGNN combines a relational GNN with disease-similarity metric learning and, against eight methods, improves indication prediction by 49.2% under zero-shot evaluation; its explainer shows multi-hop paths [2]. NBFNet represents a (source, target) pair by generalised Bellman–Ford message passing from the source, which makes it inductive and explainable by paths [10], in the spirit of subgraph reasoning [25]; A*Net and ULTRA scale and generalise the idea [26], [27]. BioPathNet adapts NBFNet to biomedical KGs with a background regulatory graph and stringent negative sampling; on TxGNN's zero-shot splits it reports indication MRR from about 0.2 (cardiovascular) to above 0.6 (cell proliferation, adrenal gland) [9].

Rare diseases and few-shot repurposing. Rare-disease studies build disease-specific KGs: literature graphs for rare neuro-muscular disorders with an ensemble of rule-based, GNN and path models [28], explainable GNNs for single rare diseases [29], semantically valid negative samples for Huntington's disease [30], and literature-based embeddings [31]. Yang et al. cast repurposing as few-shot link prediction with meta-learning [32]. We instead evaluate a zero-shot-trained model when a few known drugs are revealed, against collaborative filtering [33]. Table II positions this work.

Leakage. Leakage is a main cause of irreproducible ML-based science [34]. In KG link prediction it appears as test edges left in the graph, inverse or duplicate relations and near-duplicate entities. Random negatives add a further problem: node popularity alone separates the classes, so AUROC is uninformative when positives are rare [35], [36]. Brière et al. showed for biomedical KG embeddings that removing train–test redundancy lowers performance and that random and cold-start splits overestimate repurposing performance on Orphanet rare-disease indications; their permutation tests did not find that degree alone drives those models [37]. Our audit concerns a different failure, random negatives in evaluation, under which a degree score alone suffices.

III. Dataset and Code

PrimeKG. We use PrimeKG [11], a precision-medicine KG built from 20 resources, downloaded from Harvard Dataverse (doi:10.7910/DVN/IXA7BM; file kg.csv). Table III gives its size. Its 27 non-therapeutic relation types (protein–protein, drug–protein, disease–protein, disease–phenotype, anatomy–protein expression, drug–drug, pathways, exposures and others) form the context graph. PrimeKG has three drug–disease relations: indication (the drug is labelled for the disease), off-label use and contraindication. We call indications and off-label uses known drugs and use "approved" only for regulatory approvals we checked ourselves.

Preprocessing. PrimeKG's edge file lists every edge in both directions. For each relation we fix one canonical direction (by node type, or by node index within one type), drop duplicates and only then add an explicit inverse, so no edge enters the graph twice; with the three drug–disease relations this gives 60 relation types. Drug–disease pairs are stored in a separate table and never become context edges; for each split the message graph receives

TABLE II
Positioning against related work. Numbers are from the cited papers and use different splits and negatives; they are context, not a head-to-head comparison.

Method	KG	Unit held out	Evaluation	Reported result
Hetionet [5]	Hetionet	none (edge-level)	metapath classifier	–
DREAMwalk [22]	multi-layer KG	none (edge-level)	association prediction	up to 16.8% accuracy gain over link-prediction models
TxGNN [2]	PrimeKG	disease area	zero-shot, 8 baselines	49.2% better indication prediction
BioPathNet [9]	PrimeKG	TxGNN disease areas	MRR, Hits@k over all drugs	MRR ≈0.2–0.6+ by area
Few-shot attention [32]	medical KG	–	few-shot link prediction (meta-learning)	–
This work	PrimeKG	disease + all related diseases; autoimmune area	MRR, Hits@k, AUROC over all 7,957 drugs; zero- and few-shot; paired tests	test MRR 0.501, 68 diseases

TABLE III
PrimeKG after preprocessing, and split statistics.

Rows in kg.csv (both edge directions)	8,100,498
Nodes	129,375
drugs / diseases	7,957 / 17,080
Context edges (deduplicated)	4,007,433
relation types (with inverses)	27 (54)
Drug–disease pairs	42,631
indication / contraind. / off-label	9,388 / 30,675 / 2,568
Targets: train / validation diseases	1,206 / 68
Test split: train / test diseases	1,065 / 68
No-autoimmune: train / valid. / area	1,194 / 66 / 21
Transfer: A / A′ / C_E; test set E	11 / 11 / 1,185; 12

only the pairs of training diseases. The path channel uses no node features or text embeddings; nodes differ only by their edges. The build takes about 35 s on a laptop (bpp/build_primekg.py) and produces a 9 MB dataset used in every run.

Labels. The model predicts indications, and validation and test positives are indications only. The rare targets have so few that their labels are indications plus off-label uses: PBC 1+0, SSc 3+2, Sjögren 2+10, AIH 0+1, MG 4+3 and Crohn 16+8. Contraindications are never positives. The targets are therefore case studies; quantitative claims rest on the 68 validation, 68 test and 21 area diseases.

Code and reproducibility. All code, splits, the audit script, run logs and the analysis script that generates every number, table and figure of this paper are at https://github.com/SaiYatin/Capstone_Drug_Repurposing_Project (branch bpp-v3). The paper's numbers are LaTeX macros written by bpp/analyze.py from the run logs; none is typed by hand.

IV. Why a Strict Protocol: Audit of a Naive Pipeline

Our earlier pipeline built a graph from an enriched DRKG extract (84,712 nodes), trained an NBFNet-style model and reported test AUPRC between 0.935 and 0.999. A stand-alone audit script, released with the code, found three mechanisms behind these scores. (i) Target leakage: the message graph was built from the full KG, so all 277 test pairs were edges the model could traverse. (ii) Trivial negatives: negatives were random drug–disease pairs, and a zero-parameter score $\log \deg(\text{drug}) + \log \deg(\text{disease})$ reached test AUROC 0.972, above the model's 0.941. (iii) One-directional subgraphs: without inverse relations, five of six targets were unreachable and scored by drug features alone. In addition, 90–91% of positive labels were text-mined edges (one of them, thalidomide for PBC, contradicts a negative randomised trial [38]), a phenotype parser created 2,325 spurious relation types, language-model features were computed from identifier strings, and ablation scripts loaded the wrong model. Every number below comes from the rebuilt pipeline.

V. Zero- and Few-Shot Protocol

A. Strict-family hold-out

Holding out a disease node alone is not enough: subtypes and synonyms linked by disease–disease edges often carry the same drugs. Without further removal all 16 Crohn indications stayed reachable through "Crohn's colitis", MG through "adult-onset myasthenia gravis" and SSc through "scleroderma". A disease's family is therefore the disease plus its disease–disease neighbours; for the targets and RA/SLE/MS it also contains the disease nodes that TxGNN's name-based grouping assigns to the same group, which catches numbered subtypes. We hold out whole families:

- no drug–disease edge of any held-out family member is ever in the message graph;
- validation diseases (5% of labelled diseases, seed 0) are held out with their families, so model selection cannot reward sibling copying;
- in every training step the drug edges of the batch's query diseases and their families are also removed, so training matches zero-shot inference.

The rules live in the graph builder, and unit tests check that removed edges are absent from the propagation matrices and that no training and evaluation families overlap.

B. Selection-free test split

Validation diseases select the stopping epoch, so their scores are slightly optimistic. We therefore draw 68 further non-autoimmune labelled diseases (seed 1) as a test set, held out with their families (1,065 training diseases remain) and used neither for training nor for early stopping.

C. No-autoimmune protocol and transfer conditions

To test whether related autoimmune data matters, a second protocol also removes TxGNN's autoimmune area (all descendants of DOID:417 in the Human Disease Ontology, mapped to PrimeKG through MONDO cross-references; 70 nodes) [2] and the families of all nine autoimmune diseases studied here, 123 nodes in total. Its 21 held-out area diseases with indications form a further test set. The transfer curve uses a fixed test set E: the 12 area diseases with indications that are neither targets nor RA/SLE/MS relatives. E, the targets and the validation diseases are held out in every condition; only the training diseases change:

- A: the RA/SLE/MS families (11 diseases, 161 indications);
- A′: three sets of 11 non-autoimmune diseases, each matched to A disease by disease on number of indications;
- D: all 1,194 non-autoimmune diseases;
- C_E: all 1,185 diseases except E.

A vs. A′ isolates disease type at equal data size; C_E vs. D measures autoimmune data on top of broad data. Since the message graph contains only training diseases' drug edges, each condition also fixes what drug knowledge the model sees. An "all autoimmune" condition is impossible: beyond RA/SLE/MS, PrimeKG's autoimmune diseases are the evaluation area itself.

D. Few-shot evaluation

For each test disease with at least four indications (34 of 68) we draw a random order of its indications (3 draws). The first $k_{\max} = 3$ drugs form the reveal pool, and the remaining indications are the evaluation positives, identical for every k, so $k = 0, \ldots, 3$ are compared on the same targets. With k revealed drugs, the trained model receives the k (disease, drug) indication edges in its message graph for this query only; it is not retrained. Revealed drugs are removed from the ranking. Baselines see the same drugs: co-indication, an item-based collaborative filter [33] that scores a drug by the cosine similarity of its indication profile over training diseases to those of the revealed drugs; target similarity, the Jaccard similarity of protein-target sets; disease kNN, which ignores the revealed drugs; and disease kNN plus co-indication. We also fuse the model with co-indication by reciprocal-rank fusion (RRF, constant 60) [39], which has no tuned weight.

VI. Method

BioPathNet++ scores a query (disease u, relation indication) against every drug with two channels whose logits are added: a path channel that follows BioPathNet's NBFNet design and a prior channel of four leakage-free features. Fig. 1 shows the architecture with all layer sizes; the subsections below describe each part.

A. Path channel

For a query (u, r_q), NBFNet [10] computes a representation of every node v that summarises all paths from u to v. Node states start from an indicator boundary condition, $h_v^{(0)} = \mathbb{1}[v = u]\, q$, with $q = \mathrm{emb}(r_q) \in \mathbb{R}^d$, and each of L layers aggregates relation-specific messages:

$$m_v^{(t)} = \frac{1}{\deg v} \sum_{(x,r,v)\in\mathcal{E}} h_x^{(t-1)} \odot w_{r_q,r}^{(t)} \;+\; \mathbb{1}[v = u]\, q, \tag{1}$$

$$h_v^{(t)} = h_v^{(t-1)} + \mathrm{ReLU}\Big(\mathrm{LN}\big(W^{(t)}\big[h_v^{(t-1)};\, m_v^{(t)};\, a_v\, m_v^{(t)}\big]\big)\Big). \tag{2}$$

Here $w_{r_q,r}^{(t)}$ are query-dependent DistMult relation vectors [15], $a_v = \log(\deg v + 1)/\overline{\log(\deg +1)}$ is a PNA-style degree amplifier [40] and LN is layer normalisation. The path logit of drug d is $s_{\mathrm{path}}(u, d) = \mathrm{MLP}([h_d^{(1)}; \ldots; h_d^{(L)}; q])$. Because propagation starts at u, one forward pass scores all 7,957 drugs, and a disease whose drugs were never seen is handled like any other node.

B. Memory-efficient full-graph propagation

NBFNet implementations usually materialise per-edge messages, which needs $O(|\mathcal{E}|\, B\, d)$ memory for B queries: 4 M edges × 8 queries × 32 dimensions per layer here. We write (1) as $\sum_r (A_r H) \odot w_r$ over 60 sparse adjacency matrices A_r. A custom autograd function stores only H and recomputes $A_r H$ in the backward pass, reducing activation memory to $O(|\mathcal{V}|\, B\, d)$. The graph takes 0.32 GB; training with $B = 8$ peaks at 5.8 GB and takes about 8.5 s per step on an NVIDIA T4, and the gradients pass a finite-difference check.

C. Prior channel

The prior channel adds four features $\phi(u, d)$ per pair: (i) popularity, the number of training diseases indicated for d; (ii) a disease-kNN score, $\sum_{u'\in\mathcal{N}_k(u)} \mathrm{sim}(u, u')\, y_{u'd}$ with $k = 20$ and cosine similarity of TF–IDF-weighted disease–protein and disease–phenotype profiles; (iii) the number of d's protein targets associated with u; and (iv) the frequency of d among autoimmune training diseases. Each feature is computed with u's family excluded, identically in training and testing. A two-layer MLP g maps ϕ to a logit: $s(u, d) = s_{\mathrm{path}}(u, d) + g(\phi(u, d))$.

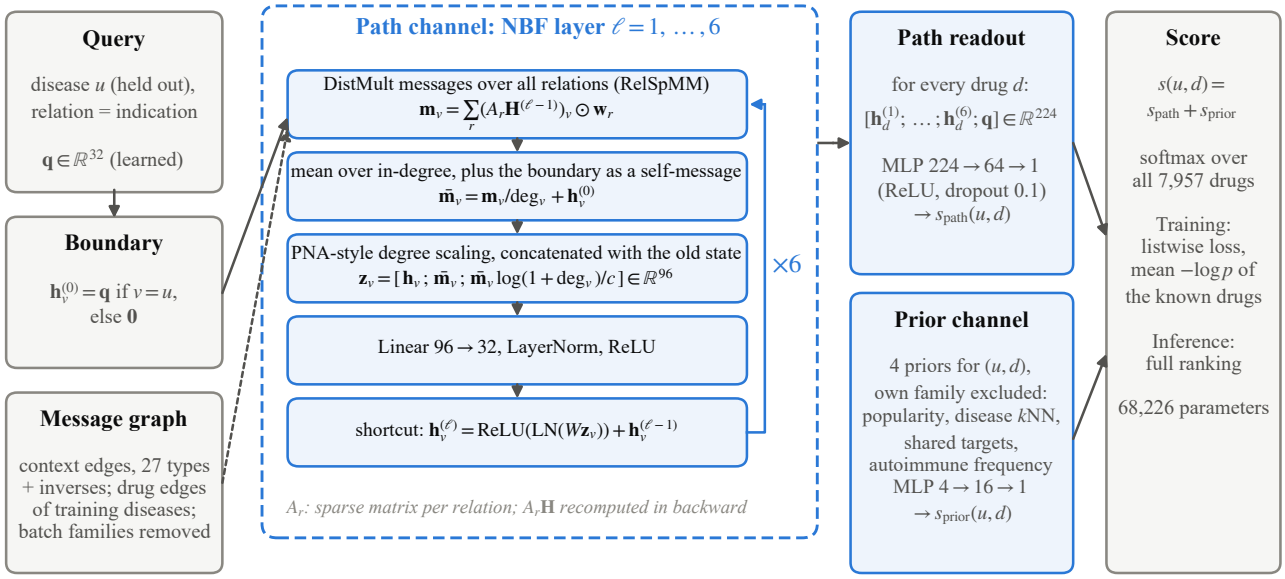


Fig. 1. BioPathNet++ architecture. A query starts as an indicator vector on disease u. Six NBF layers pass DistMult messages over a message graph that never contains the drug edges of u's family (dashed arrow: one sparse matrix A_r per relation). An MLP reads each drug's layer states into a path score, a second MLP turns four leakage-free priors into a prior score, and their sum is ranked over all 7,957 drugs.

D. Objective

Let P_u be the training indications of u. We minimise a multi-positive softmax cross-entropy over all candidates,

$$\mathcal{L}(u) = -\frac{1}{|P_u|} \sum_{d \in P_u} \log \frac{\exp s(u,d)}{\sum_{d'} \exp s(u,d')}, \tag{3}$$

so every non-indicated drug acts as a negative and no negative sampling is needed. NBFNet's self-adversarial binary cross-entropy is evaluated as an ablation.

VII. Experimental Setup

A. Baselines

All baselines use the same protocol and family exclusions. Random; drug degree; popularity (number of training diseases treated); autoimmune frequency; shared targets (drug targets among the disease's genes); and disease kNN, the drugs of the 20 most similar training diseases weighted by similarity, a transparent form of guilt by association [22]. KG embeddings: TransE, DistMult and RotatE [14]–[16], trained on exactly the message graph of the test split (all context edges plus the drug–disease edges of training diseases), with self-adversarial negative sampling [16] (256 negatives shared within each batch of 1,024 triples), 128 real dimensions per entity, Adam (learning rate 10^{-3}), 20 epochs and the epoch chosen by validation MRR (checked every 4 epochs; it peaked between epochs 12 and 20 and never exceeded 0.06); a test disease keeps its context edges, so it has an embedding even without drug edges. These settings were fixed in advance (bpp/kge.py). TxGNN is compared on its own split (Section VIII-E).

B. Metrics

For disease u with positives P_u, each positive d is ranked among all drugs after removing the other positives (filtered rank [14]):

$$\mathrm{MRR}_u = \frac{1}{|P_u|} \sum_{d \in P_u} \frac{1}{\mathrm{rank}_u(d)}, \tag{4}$$

$$\mathrm{H@}k_u = \frac{1}{|P_u|} \sum_{d \in P_u} \mathbb{1}[\mathrm{rank}_u(d) \le k], \tag{5}$$

macro-averaged over diseases [41]. We also report Any@10, the share of diseases with at least one known drug in the top 10; AUROC and AUPRC over all drugs; AUROC against hard negatives (drugs indicated for other autoimmune diseases); and AUROC against degree-matched negatives. With about 7 positives among 7,957 drugs, random ranking gives MRR and AUPRC of about 0.001 and Any@10 of 0.8%. AUROC hardly reflects this imbalance [35], so it is not a headline metric (Section VIII-A). MRR is easiest to read through ranks: BioPathNet++'s test MRR of 0.501 corresponds to a median known-drug rank of 2, with 43% of known drugs ranked first.

C. Statistics and training

Every comparison uses per-disease metrics averaged over seeds and then over diseases. We report seed mean ± sd, percentile bootstrap 95% CIs over diseases (1,000 resamples) [42] and, for comparisons, the paired bootstrap CI of the mean difference (5,000 resamples) with a two-sided Wilcoxon signed-rank test [43]. A difference is significant only if both agree at the 5% level. Training: hidden size 32, 6 layers, MLP hidden size 64, dropout 0.1, Adam with learning rate 5×10^{-3}, 8 query diseases

per batch, gradient clipping at 1.0, at most 20 epochs, early stopping on validation MRR (patience 5–6); 68,226 parameters; seeds 0–2. Transfer runs train on 11 diseases (2 steps per epoch) and may run 60 epochs with patience 10. Settings were fixed in advance, not tuned. Runs used two NVIDIA T4 GPUs on Kaggle.

VIII. Results

A. Accuracy on unseen diseases

Table IV gives every metric for the selection-free test split, the validation diseases and the targets. On the test split, a known indication is in the top 10 for 76% of diseases (disease kNN 57%, popularity 19%, chance 0.8%), and AUPRC is 0.556 against a prevalence of about 0.001. MRR is 0.501 (3 seeds) against 0.392 for disease kNN (paired difference +0.109, CI +0.044 to +0.180; Wilcoxon $p=0.003$). The KG embeddings, trained on the same graph, reach MRR 0.037 (TransE), 0.047 (DistMult) and 0.071 (RotatE), far below both. Their AUROC is high (0.878 for RotatE), but they rarely place a known drug near the top, because a held-out disease's embedding is learned only from its non-drug edges (genes, phenotypes, related diseases), with no drug edge to anchor it.

On the 68 validation diseases (Fig. 2), BioPathNet++ reaches MRR 0.394±0.003 against 0.281 for disease kNN and 0.046 for popularity (paired difference +0.113, CI +0.056 to +0.179; $p<0.001$); it is better on 43 diseases, tied on 13 and worse on 12. Against hard negatives AUROC is 0.799, versus 0.661 for disease kNN and 0.476 (below chance) for popularity, so the model does more than rank generically popular drugs. On the six targets MRR is 0.475 against 0.296; the gain points the same way on 4 of 6 targets but is not significant with six diseases ($p=0.188$). BioPathNet++ therefore outperforms every baseline evaluated on the same leakage-free split. The test diseases are non-autoimmune and easier for every method (disease kNN rises from 0.281 to 0.392), so test and validation values are not comparable with each other.

Why AUROC is not the headline metric. Baselines without learning reach validation AUROC 0.78–0.86, and popularity 0.830, yet their MRR is below 0.06. With thousands of negatives per positive, AUROC rewards placing positives above the long tail of irrelevant drugs and says little about the top of the list, which is what a clinician reads. The same effect let a degree heuristic reach AUROC 0.972 in the audit.

B. Transfer to rare autoimmune diseases

Fig. 3 and Table V give the transfer curve. Scarce data. Trained only on the RA/SLE/MS families (A), BioPathNet++ reaches MRR 0.290 on E, against 0.126 with label-matched non-autoimmune diseases (A′): a significant difference of +0.164 (CI +0.037 to +0.316; $p=0.016$; better on 9 of 12). It also holds for final instead of early-stopped checkpoints (+0.146, $p=0.021$), and the rare targets point the same way (0.124 vs. 0.041; $p=0.062$, $n=6$). Disease kNN shows the same pattern (0.258 vs. 0.086), and the model is only slightly ahead of it. With 11 training diseases the transferable knowledge is a drug-class prior: the RA/SLE/MS families are treated mostly with corticosteroids and TNF inhibitors, and the A-trained model ranks the same corticosteroids first for every target. With paths only (3 seeds), A and A′ both stay near the floor (0.007 vs. 0.010), so the scarce-data advantage comes from the prior channel.

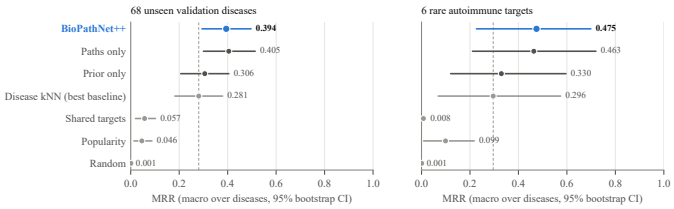


Fig. 2. Validation MRR with 95% bootstrap CIs over diseases (seed-averaged). Dashed line: disease kNN.

Full scale. Training on all non-autoimmune diseases (D) raises MRR on E to 0.532; adding the autoimmune diseases (C_E, n=3) does not help but lowers it to 0.410 (C_E minus D: −0.122, CI −0.200 to −0.052, $p=0.010$; lower on 9 of 12 diseases). Both conditions stop on the same validation diseases, and disease kNN drops in the same way (0.318 to 0.271), so the drop comes from which diseases are in the training set rather than from the model or from early stopping. On the targets, removing all 123 autoimmune nodes leaves MRR unchanged (0.487 vs. 0.475; 3 seeds; $p=0.812$), and on the area diseases the model reaches 0.449 against 0.311 for disease kNN (CI +0.010 to +0.280, $p=0.147$; suggestive by our rule). The best source thus depends on data size: related autoimmune diseases when data is scarce, the broad pharmacological graph at full scale.

C. Few-shot: a few known drugs

Table VI shows what a few known drugs add on the 34 test diseases with at least four indications. Revealed as graph edges, they help the zero-shot-trained model without retraining: MRR rises from 0.313 to 0.362, 0.419 and 0.533 for one, two and three revealed drugs, a significant gain for every k (at $k=3$: +0.220, CI +0.147 to +0.296; $p<0.001$). The model stays significantly ahead of disease kNN at every k ($p=0.039$ at $k=3$). Co-indication filtering, which recommends drugs indicated for the same training diseases as the revealed ones, catches up quickly: the model's lead is not significant from $k=1$ (+0.065, $p=0.153$), and at $k=3$ the two are tied (−0.006, $p=0.792$). Once some drugs are known, much of the signal is therefore "drugs used together", which collaborative filtering captures directly, whereas the model, trained only zero-shot, sees the revealed drugs merely as extra edges. The two views are complementary. Reciprocal-rank fusion of the model with co-indication gives the highest MRR at every $k \geq 1$ (0.384, 0.476, 0.607), significantly above the model alone at $k=3$ ($p=0.010$), and +0.033 above the best baseline, disease kNN plus co-indication, at $k=3$

TABLE IV
Zero-shot drug ranking over all 7,957 drugs. Test: 68 diseases never used for training or early stopping (3 seeds; KG embeddings 3 seeds each). Any@10: diseases with a known indication in the top 10 (random expectation 0.8%). H@k: known drugs in the top k. $\text{AUROC}_{\text{hard}}$: known drugs vs. drugs of other autoimmune diseases; $\text{AUROC}_{\text{deg}}$: vs. degree-matched drugs. Random AUPRC ≈ prevalence (≈0.001).

Set	Method	Any@10	H@1	H@10	H@50	AUROC	$\text{AUROC}_{\text{hard}}$	$\text{AUROC}_{\text{deg}}$	AUPRC	MRR	Mean rank
Test (68, never selected)	Random	1%	0.000	0.005	0.006	0.453	0.471	0.500	0.002	0.001	4,349
	Popularity	19%	0.042	0.070	0.267	0.885	0.562	0.812	0.084	0.062	918
	TransE	27%	0.019	0.061	0.146	0.880	0.674	0.773	0.050	0.037	952
	DistMult	22%	0.025	0.089	0.193	0.852	0.638	0.732	0.060	0.047	1,175
	RotatE	27%	0.052	0.094	0.209	0.878	0.723	0.785	0.089	0.071	971
	Disease kNN	57%	0.347	0.459	0.583	0.907	0.721	0.859	0.429	0.392	742
	BioPathNet++	76%	0.435	0.637	0.727	0.959	0.862	0.924	0.556	0.501	329
Validation (68)	Random	1%	0.000	0.002	0.004	0.471	0.488	0.490	0.002	0.001	4,205
	Popularity	15%	0.034	0.049	0.188	0.830	0.476	0.760	0.069	0.046	1,355
	Disease kNN	44%	0.239	0.350	0.490	0.860	0.661	0.822	0.313	0.281	1,118
	BioPathNet++	56%	0.353	0.456	0.601	0.938	0.799	0.886	0.424	0.394	495
Rare targets (6)	Random	0%	0.000	0.000	0.000	0.476	0.488	0.389	0.002	0.001	4,165
	Popularity	50%	0.062	0.148	0.162	0.905	0.460	0.766	0.116	0.099	755
	Disease kNN	67%	0.243	0.424	0.725	0.962	0.778	0.917	0.407	0.296	299
	BioPathNet++	83%	0.396	0.648	0.695	0.982	0.892	0.954	0.594	0.475	141

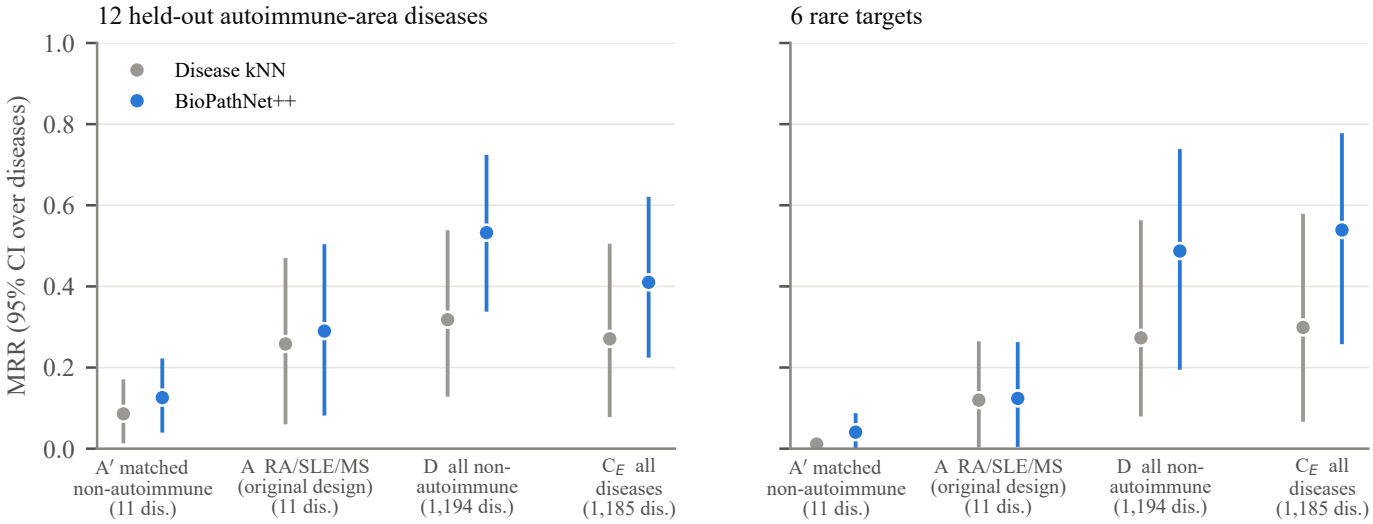


Fig. 3. Transfer curve. MRR on the 12 held-out autoimmune diseases E (left) and the six rare targets (right) per training condition, BioPathNet++ vs. disease kNN, with 95% bootstrap CIs over diseases.

TABLE V
Transfer curve: MRR [95% CI] by training condition. Test diseases are held out in every condition.

		12 held-out autoimmune-area diseases (E)				6 rare targets			
Training data	#dis.	BioPathNet++		disease kNN		BioPathNet++		disease kNN	
A′: 11 matched non-autoimmune diseases	11	0.126	[0.04, 0.22]	0.086	[0.02, 0.17]	0.041	[0.00, 0.08]	0.011	[0.00, 0.02]
A: RA/SLE/MS families (11 diseases)	11	0.290	[0.09, 0.50]	0.258	[0.06, 0.47]	0.124	[0.01, 0.26]	0.120	[0.01, 0.26]
D: all non-autoimmune diseases	1,194	0.532	[0.34, 0.72]	0.318	[0.13, 0.53]	0.487	[0.20, 0.73]	0.273	[0.08, 0.56]
C_E: all diseases except the test set	1,185	0.410	[0.23, 0.62]	0.271	[0.08, 0.50]	0.539	[0.26, 0.77]	0.299	[0.07, 0.57]

TABLE VI
Few-shot ranking on the 34 test diseases with ≥4 indications: MRR on the same held-back indications after revealing k known drugs (3 random draws; BioPathNet++ averaged over 3 seeds, not retrained). Best per column in bold.

Revealed known drugs k	MRR, k=0	MRR, k=1	MRR, k=2	MRR, k=3	H@10, k=3
BioPathNet++ (graph edges, no retraining)	0.313	0.362	0.419	0.533	0.709
BioPathNet++ + co-indication (RRF)	0.313	0.384	0.476	0.607	0.767
Disease kNN (ignores revealed drugs)	0.231	0.261	0.310	0.412	0.494
Co-indication (item-based CF)	0.080	0.297	0.416	0.539	0.673
Target similarity (Jaccard)	0.080	0.032	0.046	0.041	0.114
Disease kNN + co-indication	0.227	0.347	0.443	0.574	0.729

(CI −0.021 to +0.080, p=0.082), a margin that is not significant with 34 diseases. Target similarity is weak at every k. Training the model with randomly revealed drugs is the natural next step.

D. What each component contributes

Table VII isolates the components on the validation diseases; it lists the number of seeds per variant. Without the prior channel (paths only), MRR is 0.405, tied with the full model (full minus paths only: −0.010, p=0.148). Without path reasoning (prior only), MRR falls to 0.306 (p<0.001), close to disease kNN, one of its inputs. Path reasoning is therefore the source of the gain; the prior channel reduces seed variance on the targets (sd 0.087 to 0.017), and in the no-autoimmune protocol it adds +0.033 MRR on the 21 area diseases, not significantly (CI −0.016 to +0.084; p=0.227; 3 seeds each). With NBFNet's self-adversarial BCE instead of the listwise softmax, MRR is 0.378, not significantly lower (full minus BCE: +0.016, CI

−0.005 to +0.039; p =0.344); BCE converges more slowly (two of its three seeds were still improving at the 20-epoch cap) and is lower on the six targets (0.320 vs. 0.475, too few to test). We keep the listwise loss because it needs no negative sampling. Following BioPathNet's background graph [9], dropping tissue-expression and drug–drug edges shrinks the graph from 4,007,433 to 1,133,029 edges (72% fewer) at unchanged MRR (0.404; p =0.299), with 11% faster steps and lower memory (5.52 vs. 5.76 GB). Validation diseases also select the epoch; from the training logs this adds 0.002–0.016 MRR of optimism for the full model, far below its gain over disease kNN, and the selection-free test split confirms that gain.

E. Head-to-head with TxGNN on its own split

We ran both models on TxGNN's autoimmune disease-area split, generated by TxGNN's code (commit 0e0203c; Table VIII). Its random test edges are drawn without a seed, so we fixed the seed and the draw differs from the published runs. The test set is the 13 held-out area diseases with indications (75 pairs). We converted the split so that our model never sees an edge that TxGNN does not train on: the 3,248 random test edges of 851 other diseases are removed from our graph and labels, and area diseases are held out with their families. BioPathNet++ trains on 1,139 diseases (63 more for early stopping), while TxGNN learns from all 1,299 non-test diseases with indications, so our model still sees less. TxGNN used its reproduction settings (one pre-training epoch, 500 fine-tuning epochs); each model ran 3 seeds. Our implementation of TxGNN's protocol reproduces TxGNN's own evaluation exactly for every seed.

Over all 7,957 drugs, BioPathNet++ is significantly better: MRR 0.273 against 0.048 (+0.226, CI +0.055 to +0.422, p =0.040), H@100 0.618 against 0.232 (p =0.002) and AUROC 0.936 against 0.753 (p <0.001). Under TxGNN's protocol, which ranks only the 1,798 drugs with at least one indication, no difference is significant (Recall@100 0.614 vs. 0.536; AUROC 0.777 vs. 0.901, p =0.168). The ranked lists explain why: only 23% of TxGNN's top-100 drugs have any indication, against 99% for BioPathNet++, and TxGNN's protocol removes those drugs before scoring. In a first, stricter conversion our model trained on only 637 diseases; full coverage raised its full-ranking MRR by +0.213 (p =0.013), while the AUROC difference under TxGNN's protocol (ours minus TxGNN) changed only from −0.163 to −0.124. The comparison rests on one split and 13 diseases.

F. Sparsity

Rare diseases are sparsely annotated, so we stratify the validation diseases by number of context edges (Fig. 4). Accuracy grows with connectivity (Spearman ρ = 0.30, p =0.012), and BioPathNet++ is ahead of disease kNN in every tertile: 0.236 vs. 0.160 (sparsest), 0.389 vs. 0.218 and 0.557 vs. 0.462 (densest).

TABLE VII
Ablations on the 68 validation diseases (means over n seeds). p: two-sided Wilcoxon on paired MRR vs. the full model.

Variant (seeds)	Valid MRR	Valid H@10	Targets MRR	p vs full (valid)
Full model (n=3)	0.394±0.003	0.456±0.018	0.475±0.017	–
− prior channel (paths only) (n=3)	0.405±0.013	0.479±0.025	0.463±0.087	0.148
− path reasoning (prior only) (n=3)	0.306±0.003	0.354±0.006	0.330±0.025	<0.001
BRG relation subset (n=3)	0.404±0.006	0.492±0.004	0.504±0.075	0.299
BCE loss instead of listwise (n=3)	0.378±0.028	0.452±0.017	0.320±0.046	0.344

TABLE VIII
TxGNN's autoimmune split (13 test diseases), means over seeds. TxGNN protocol: only drugs with ≥1 indication are ranked; full ranking: all 7,957 drugs. Lower block: first run with stricter coverage. Published row: TxGNN's repository (different draw).

		TxGNN protocol			full ranking		
Method	seeds	R@100	AUROC	AUPRC	MRR	H@100	AUROC
TxGNN, published	5	0.338	–	–	–	–	–
Full training coverage (main comparison)							
TxGNN	3	0.536	0.901	0.161	0.048	0.232	0.753
BioPathNet++ (1,139 train dis.)	3	0.614	0.777	0.343	0.273	0.618	0.936
Strict coverage: every disease with a test edge held out (ablation)							
TxGNN (unchanged)	1	0.474	0.909	0.147	0.026	0.155	0.736
BioPathNet++ (637 train dis.)	1	0.475	0.746	0.101	0.061	0.398	0.910

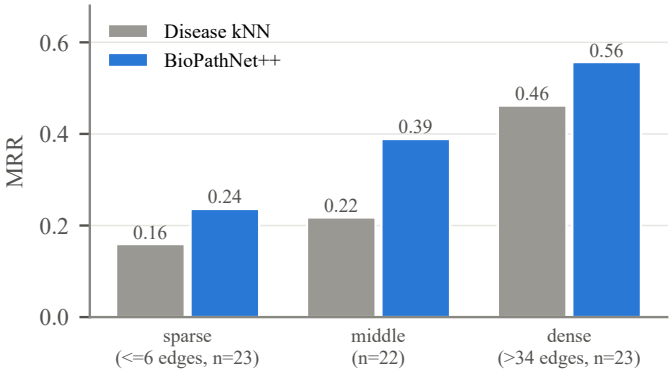


Fig. 4. Validation MRR by connectivity tertile of the held-out disease (number of context edges).

G. Case studies

Table IX and Fig. 5 show the six targets. PBC. Ursodeoxycholic acid, first-line therapy and the only labelled drug, ranks first in all seeds, followed by other bile acids. MG. The four labelled cholinesterase inhibitors (ambenonium, neostigmine, pyridostigmine, distigmine) rank 1–4 in all seeds; Table X shows the paths behind such predictions. SSc. Labelled aminobenzoic acid, benzoic acid and bosentan lead, followed by corticosteroids. Sjögren. Labelled tear-substitute components and prednisolone lead. Crohn. Labelled corticosteroids rank 1–4; many of the 24 labelled drugs rank low, which keeps MRR modest. AIH. PrimeKG lists a single off-label drug, while the model ranks prednisolone first in two of three seeds; corticosteroids are standard AIH therapy (prednisone is the comparator in NCT00838214). This is a label gap, so metrics on sparsely labelled diseases are lower bounds.

H. Validation of new candidates

Ranking metrics only reward drugs PrimeKG already labels. To judge the unlabelled candidates (each target's

TABLE IX
Per-target MRR (seed mean) and top predictions (seed 0).
* labelled in PrimeKG.

Target	#known	kNN	Paths only	BioPathNet++	Best rank	Top-3 predictions (seed 0)
PBC	1	1.000	1.000	1.000	1	Ursodeoxycholic acid*, Doripenem, Cholic Acid
SSc	5	0.041	0.437	0.452	1	Benzoic acid*, Bosentan*, Limaprost
Sjögren	12	0.152	0.731	0.510	1	Prednisolone*, Polyethylene glycol 400*, Propylene glycol*
AIH	1	0.001	0.002	0.002	>100	Prednisolone, Triamcinolone, Hydrocortisone
MG	7	0.179	0.418	0.671	1	Ambenonium*, Neostigmine*, Pyridostigmine*
Crohn	24	0.403	0.191	0.212	1	Hydrocortisone*, Triamcinolone*, Cortisone acetate*

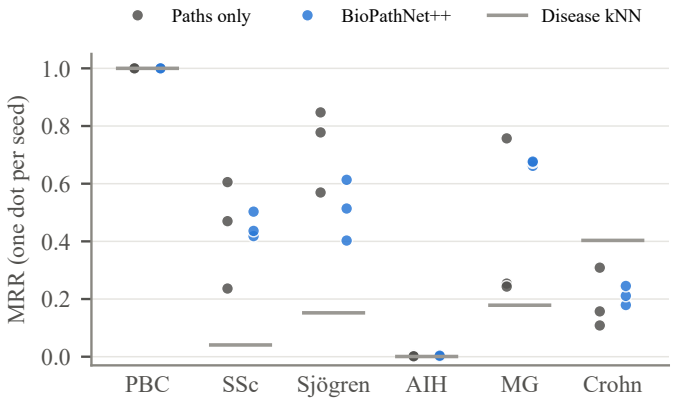


Fig. 5. Per-target MRR for each seed. The prior channel narrows the spread across seeds (SSc, Sjögren, MG).

top 20 by seed-averaged rank; 86 of 120 are unlabelled), we compare them with the popularity baseline's top 20 and 20 random drugs per target (Table XI; bpp/validate.py, all API responses cached).

Clinical trials. We searched ClinicalTrials.gov (API v2) with disease synonyms. A trial counts only if it names the drug as an intervention, matched by the whole drug name; grades are A (phase ≥3), B (phase 2) and C (phase 1 or observational). 17% of the model's candidates have such a trial (11 at phase ≥3), against 14% for popularity and 0% for random drugs. Corticosteroids often appear only as background therapy, premedication or transplant conditioning, so a stricter count keeps only trials whose title names the drug or in which it is in some but not all arms: 13% (9 at phase ≥3) against 8% and 0%. Examples are amifampridine for MG (NCT03304054), ciprofloxacin (NCT00609973) and clarithromycin (NCT00269386) for Crohn, N-acetylcysteine for SSc (NCT00428883), and ustekinumab for Crohn, FDA-approved in 2016 but unlabelled in PrimeKG; every cited trial was checked by hand. A trial is not proof of benefit: amifampridine helped in MuSK-antibody MG [44] but not, in a 2026 trial, in acetylcholine-receptor-antibody MG [45]. On the transfer model C_E, 9 of the 12 area diseases have a labelled drug in the top 10, and 21 of 74 unlabelled top-10 drugs have a registered trial, e.g. canakinumab for CINCA syndrome (NCT00685373).

Literature. For each pair we retrieved PubMed hit counts and the three most relevant titles, classified with AI assistance and then checked against the abstracts behind every supporting and against label (animal-only studies, drugs in every arm and inconclusive results count as mixed; 17 labels changed). The model's candidates are co-mentioned with their disease somewhat less often than popular drugs (63% vs. 72%), but more often with evidence of benefit (15% vs. 10%; random 0%) and less often with harm or no benefit (5% vs. 12%): three corticosteroid pairs and N-acetylcysteine for SSc, whereas popularity's are mostly antibiotics that can worsen MG or induce hepatitis.

Temporal validation. PrimeKG's drug labels predate 2022. Six drugs approved for our targets since then are in PrimeKG (Table XII); seladelpar, efgartigimod, zilucoplan, nipocalimab and mirikizumab are absent. Their median rank is 1,104 of 7,957 (top 14%) and 5 of 6 beat the random median; risankizumab ranks 37. With six drugs this is not significant (sign test $p = 0.109$), and disease kNN ranks three of the six higher.

Mechanism and paths. 17% of the model's candidates have at least one target among the disease's genes, against 16% for popularity and 3% for random drugs; a Reactome immune-pathway enrichment test found nothing significant in any set. Gradient-times-input attribution on every edge message followed by beam search [10] extracts the paths behind a prediction (Table X). Cholinesterase inhibitors reach MG through the MuSK gene and neuromuscular-junction disease, amifampridine through Lambert–Eaton myasthenic syndrome, and ursodeoxycholic acid reaches PBC through biliary diseases; ciprofloxacin, however, reaches Crohn through "mouth disease" and sinusitis, a weak link. By name, corticosteroids fill 41% and antibacterials 21% of all top-20 slots, so lists need expert filtering; a review sheet with all evidence is prepared (expert_review_sheet.xlsx).

IX. Conclusion

Path-based zero-shot transfer works for rare autoimmune diseases that a model has never seen. With a handful of training diseases, related autoimmune diseases are the most useful source; with the full graph, adding the autoimmune diseases does not help and even lowers accuracy on held-out autoimmune diseases, so broad pharmacology and biology are the better source. Under a strict-family protocol with full ranking, BioPathNet++ outperforms every baseline evaluated on the same split, including KG embeddings, and on TxGNN's own split it ranks all drugs significantly better than TxGNN. When a few drugs of a disease are known, the model improves without retraining, but co-indication filtering catches up after one or two drugs; fusing the two gives the best rankings, although not significantly better than the strongest baseline. Path reasoning drives accuracy, the prior channel adds stability, and most of PrimeKG's edges are redundant for this task.

How the numbers compare. Our MRRs lie within the range BioPathNet reports on TxGNN's disease areas [9], but those splits remove disease areas and most molecular edges whereas we hold out families and keep molecular context, so we compare with TxGNN only on a shared split. The earlier pipeline's AUPRC of 0.999 measured an easier task, separating graph edges from random

TABLE X
Highest-scoring explanation path for four predictions (seed 0); edge labels are PrimeKG relations ($^{-1}$: inverse direction).

Pyridostigmine for Myasthenia Gravis: myasthenia gravis $\xrightarrow{\text{disease protein}^{-1}}$ MUSK $\xrightarrow{\text{disease protein}}$ neuromuscular junction disease $\xrightarrow{\text{indication (disease}\to\text{drug)}}$ Pyridostigmine
Ursodeoxycholic acid for Primary Biliary Cholangitis: primary biliary cholangitis $\xrightarrow{\text{disease disease}}$ genetic biliary tract disease $\xrightarrow{\text{disease disease}}$ choledocholithiasis $\xrightarrow{\text{indication (disease}\to\text{drug)}}$ Ursodeoxycholic acid
Amifampridine for Myasthenia Gravis: myasthenia gravis $\xrightarrow{\text{disease disease}}$ immune-mediated acquired neuromuscular junction disease $\xrightarrow{\text{disease disease}^{-1}}$ Lambert-Eaton myasthenic syndrome $\xrightarrow{\text{indication (disease}\to\text{drug)}}$ Amifampridine
Ciprofloxacin for Crohn Disease: Crohn disease $\xrightarrow{\text{disease disease}^{-1}}$ mouth disease $\xrightarrow{\text{disease disease}}$ maxillary sinusitis $\xrightarrow{\text{indication (disease}\to\text{drug)}}$ Ciprofloxacin

TABLE XI
Evidence for unlabelled candidates: BioPathNet++ top-20 vs. popularity top-20 vs. random drugs (six targets).

Evidence for unlabelled candidates	BioPathNet++ top-20 (n=86)	Popularity top-20 (n=120)	Random (n=120)
Trial names the drug, grade A–C	17%	14%	0%
Trial names the drug, phase ≥2 (A–B)	15%	12%	0%
Trial tests the drug, grade A–C	13%	8%	0%
Trial tests the drug, phase ≥2 (A–B)	12%	8%	0%
Any PubMed co-mention	63%	72%	18%
PubMed: evidence of benefit	15%	10%	0%
PubMed: harm or no benefit	5%	12%	0%
≥1 drug target among disease genes	17%	16%	3%

TABLE XII
Ranks of drugs approved after the PrimeKG snapshot (top) and of earlier approvals missing from its labels (bottom); mean over three seeds.

Drug	Target	FDA approval	BioPathNet++ rank	Top %	kNN rank	Popularity rank
Risankizumab	Crohn	2022-06	37	0.5	30	951
Upadacitinib	Crohn	2023-05	550	6.9	127	1,527
Ravulizumab	MG	2022-04	845	10.6	1,383	1,381
Guselkumab	Crohn	2025-03	1,363	17.1	5,570	5,570
Rozanolixizumab	MG	2023-06	2,278	28.6	7,182	7,182
Elafibranor	PBC	2024-06	5,825	73.2	3,266	3,266
Tocilizumab	SSc	2021-03	58	0.7	35	260
Nintedanib	SSc	2019-09	848	10.7	375	365
Obeticholic acid	PBC	2016-05	2,272	28.6	6,452	6,452

pairs; near-perfect zero-shot numbers should be met with scepticism.

Limitations. (i) The targets have 1–24 labelled drugs, so per-target metrics are noisy and can be lower bounds (AIH). (ii) E has 12 diseases and the TxGNN comparison 13; BioPathNet itself was not re-run on a shared split. (iii) Corticosteroids and antibiotics dominate the lists, partly a popularity bias. (iv) PrimeKG labels are incomplete, so some true candidates count as false positives. (v) The few-shot model is not trained with revealed drugs. (vi) Literature labels were AI-assisted and checked against abstracts, but no clinician has reviewed the candidates; the rankings are hypotheses, not treatment recommendations. Future work: few-shot training, BioPathNet and TxGNN on further shared splits, and clinician review with the prepared sheet.

Data and Code Availability

PrimeKG: Harvard Dataverse, doi:10.7910/DVN/IXA7BM. Code, splits, the audit script, run logs and the analysis that generates every number, table and figure: https://github.com/SaiYatin/Capstone_Drug_Repurposing_Project (branch bpp-v3).

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Generative AI tools were used to assist with code development, data analysis and language editing. All results were verified by the authors against the released run logs.

References

[1] S. Pushpakom, F. Iorio, P. A. Eyers, K. J. Escott, S. Hopper, A. Wells et al., "Drug repurposing: progress, challenges and recommendations," Nature Reviews Drug Discovery, vol. 18, no. 1, pp. 41–58, 2019.

[2] K. Huang, P. Chandak, Q. Wang, S. Havaldar, A. Vaid, J. Leskovec, G. N. Nadkarni, B. S. Glicksberg, N. Gehlenborg, and M. Zitnik, "A foundation model for clinician-centered drug repurposing," Nature Medicine, vol. 30, no. 12, pp. 3601–3613, 2024.

[3] S. Nguengang Wakap, D. M. Lambert, A. Olry, C. Rodwell, C. Gueydan, V. Lanneau, D. Murphy, Y. Le Cam, and A. Rath, "Estimating cumulative point prevalence of rare diseases: analysis of the Orphanet database," European Journal of Human Genetics, vol. 28, no. 2, pp. 165–173, 2020.

[4] N. Conrad, S. Misra, J. Y. Verbakel, G. Verbeke, G. Molenberghs et al., "Incidence, prevalence, and co-occurrence of autoimmune disorders over time and by age, sex, and socioeconomic status: a population-based cohort study of 22 million individuals in the UK," The Lancet, vol. 401, no. 10391, pp. 1878–1890, 2023.

[5] D. S. Himmelstein, A. Lizee, C. Hessler, L. Brueggeman, S. L. Chen, D. Hadley, A. Green, P. Khankhanian, and S. E. Baranzini, "Systematic integration of biomedical knowledge prioritizes drugs for repurposing," eLife, vol. 6, p. e26726, 2017.

[6] S. Bonner, I. P. Barrett, C. Ye, R. Swiers, O. Engkvist, A. Bender, C. T. Hoyt, and W. L. Hamilton, "A review of biomedical datasets relating to drug discovery: a knowledge graph perspective," Briefings in Bioinformatics, vol. 23, no. 6, p. bbac404, 2022.

[7] S. Wei, C. Sasi, J. Piepenbrock, M. A. Huynen, and P. A. C. 't Hoen, "The use of knowledge graphs for drug repurposing: From classical machine learning algorithms to graph neural networks," Computers in Biology and Medicine, vol. 196, p. 110873, 2025.

[8] A. Kumar, S. Bhandary, S. G. Hegde, and J. Chatterjee, "Knowledge graph applications and multi-relation learning for drug repurposing: A scoping review," Computational Biology and Chemistry, vol. 115, p. 108364, 2025.

[9] E. Y. Hu, S. Oleshko, S. Firmani, H. G. Cheng, Z. Zhu, M. A. Wörheide, M. Arnold, M. Colomé-Tatché, J. Tang, S. Xhonneux, and A. Marsico, "Enhancing link prediction in biomedical knowledge graphs with BioPathNet," Nature Biomedical Engineering, vol. 10, no. 8, pp. 1747–1769, 2026.

[10] Z. Zhu, Z. Zhang, L.-P. Xhonneux, and J. Tang, "Neural Bellman-Ford networks: A general graph neural network framework for link prediction," in Advances in Neural Information Processing Systems (NeurIPS), 2021, arXiv:2106.06935.

[11] P. Chandak, K. Huang, and M. Zitnik, "Building a knowledge graph to enable precision medicine," Scientific Data, vol. 10, p. 67, 2023.

[12] V. N. Ioannidis, X. Song, S. Manchanda, M. Li, X. Pan, D. Zheng, X. Ning, X. Zeng, and G. Karypis, "DRKG – drug repurposing knowledge graph for COVID-19," https://github.com/gnn4dr/DRKG, 2020.
[13] S. Bonner, I. P. Barrett, C. Ye, R. Swiers, O. Engkvist, C. T. Hoyt, and W. L. Hamilton, "Understanding the performance of knowledge graph embeddings in drug discovery," Artificial Intelligence in the Life Sciences, vol. 2, p. 100036, 2022.
[14] A. Bordes, N. Usunier, A. Garcia-Duran, J. Weston, and O. Yakhnenko, "Translating embeddings for modeling multi-relational data," in Advances in Neural Information Processing Systems (NeurIPS), 2013.
[15] B. Yang, W.-t. Yih, X. He, J. Gao, and L. Deng, "Embedding entities and relations for learning and inference in knowledge bases," in International Conference on Learning Representations (ICLR), 2015, arXiv:1412.6575.
[16] Z. Sun, Z.-H. Deng, J.-Y. Nie, and J. Tang, "RotatE: Knowledge graph embedding by relational rotation in complex space," in International Conference on Learning Representations (ICLR), 2019, arXiv:1902.10197.
[17] M. Schlichtkrull, T. N. Kipf, P. Bloem, R. van den Berg, I. Titov, and M. Welling, "Modeling relational data with graph convolutional networks," in The Semantic Web (ESWC), ser. Lecture Notes in Computer Science, vol. 10843, 2018, pp. 593–607.
[18] R. Zhang, D. Hristovski, D. Schutte, A. Kastrin, M. Fiszman, and H. Kilicoglu, "Drug repurposing for COVID-19 via knowledge graph completion," Journal of Biomedical Informatics, vol. 115, p. 103696, 2021.
[19] D. Morselli Gysi, Í. do Valle, M. Zitnik, A. Ameli, X. Gan et al., "Network medicine framework for identifying drug-repurposing opportunities for COVID-19," Proceedings of the National Academy of Sciences, vol. 118, no. 19, p. e2025581118, 2021.
[20] M. Zitnik, M. Agrawal, and J. Leskovec, "Modeling polypharmacy side effects with graph convolutional networks," Bioinformatics, vol. 34, no. 13, pp. i457–i466, 2018.
[21] Z. Gao, P. Ding, and R. Xu, "KG-Predict: A knowledge graph computational framework for drug repurposing," Journal of Biomedical Informatics, vol. 132, p. 104133, 2022.
[22] D. Bang, S. Lim, S. Lee, and S. Kim, "Biomedical knowledge graph learning for drug repurposing by extending guilt-by-association to multiple layers," Nature Communications, vol. 14, p. 3570, 2023.
[23] M. Ali, M. Berrendorf, C. T. Hoyt, L. Vermue, M. Galkin, S. Sharifzadeh, A. Fischer, V. Tresp, and J. Lehmann, "Bringing light into the dark: A large-scale evaluation of knowledge graph embedding models under a unified framework," IEEE Transactions on Pattern Analysis and Machine Intelligence, vol. 44, no. 12, pp. 8825–8845, 2022.
[24] Ç. Oğuztüzün, Z. Gao, H. Li, and R. Xu, "KGiA: Drug repurposing through disease-aware knowledge graph augmentation," Journal of Biomedical Informatics, vol. 168, p. 104857, 2025.
[25] K. K. Teru, E. Denis, and W. L. Hamilton, "Inductive relation prediction by subgraph reasoning," in International Conference on Machine Learning (ICML), 2020.
[26] Z. Zhu, X. Yuan, M. Galkin, L.-P. Xhonneux, M. Zhang, M. Gazeau, and J. Tang, "A*Net: A scalable path-based reasoning approach for knowledge graphs," in Advances in Neural Information Processing Systems (NeurIPS), 2023.
[27] M. Galkin, X. Yuan, H. Mostafa, J. Tang, and Z. Zhu, "Towards foundation models for knowledge graph reasoning," in International Conference on Learning Representations (ICLR), 2024, arXiv:2310.04562.
[28] F. Papadimas, S. Svolou, K. Bougiatiotis, F. Aisopos, A. Krithara et al., "Identifying drug repurposing candidates for rare neuro-muscular disorders, using different AI methods on the literature knowledge graph," JAMIA Open, vol. 9, no. 2, 2026.
[29] P. Perdomo-Quinteiro, K. Wolstencroft, M. Roos, and N. Queralt-Rosiñach, "Knowledge graphs and explainable artificial intelligence (AI) for drug repurposing on rare diseases," Neurosymbolic Artificial Intelligence, vol. 2, 2026.
[30] N. Bianchi, A. Lefebvre, K. Wolstencroft, and M. R. Spruit, "Design and evaluation of semantically-valid negative samples integration techniques for scalable semi-automated drug repurposing prediction pipelines in rare disease research," BMC Bioinformatics, vol. 27, no. 1, 2026.
[31] D. N. Sosa, A. Derry, M. Guo, E. Wei, C. Brinton, and R. B. Altman, "A literature-based knowledge graph embedding method for identifying drug repurposing opportunities in rare diseases," Pacific Symposium on Biocomputing, vol. 25, pp. 463–474, 2020.
[32] C. Yang, X. Chen, J. Huang, Y. An, Z. Huang, and Y. Sun, "A few-shot link prediction framework to drug repurposing using multi-level attention network," Computers in Biology and Medicine, vol. 170, p. 107936, 2024.
[33] B. Sarwar, G. Karypis, J. Konstan, and J. Riedl, "Item-based collaborative filtering recommendation algorithms," in Proceedings of the 10th International Conference on World Wide Web (WWW), 2001, pp. 285–295.
[34] S. Kapoor and A. Narayanan, "Leakage and the reproducibility crisis in machine-learning-based science," Patterns, vol. 4, no. 9, p. 100804, 2023.
[35] T. Saito and M. Rehmsmeier, "The precision-recall plot is more informative than the ROC plot when evaluating binary classifiers on imbalanced datasets," PLoS ONE, vol. 10, no. 3, p. e0118432, 2015.
[36] J. Davis and M. Goadrich, "The relationship between precision-recall and ROC curves," in Proceedings of the 23rd International Conference on Machine Learning (ICML), 2006, pp. 233–240.
[37] G. Brière, T. Stosskopf, B. Loire, and A. Baudot, "Benchmarking the impact of data leakage on the performance of knowledge graph embedding models for biomedical link prediction," Bioinformatics, vol. 42, no. 8, p. btag608, 2026.
[38] P. A. McCormick, F. Scott, O. Epstein, A. K. Burroughs, P. J. Scheuer, and N. McIntyre, "Thalidomide as therapy for primary biliary cirrhosis: a double-blind placebo controlled pilot study," Journal of Hepatology, vol. 21, no. 4, pp. 496–499, 1994.
[39] G. V. Cormack, C. L. A. Clarke, and S. Büttcher, "Reciprocal rank fusion outperforms Condorcet and individual rank learning methods," in Proceedings of the 32nd International ACM SIGIR Conference on Research and Development in Information Retrieval, 2009, pp. 758–759.
[40] G. Corso, L. Cavalleri, D. Beaini, P. Liò, and P. Veličković, "Principal neighbourhood aggregation for graph nets," in Advances in Neural Information Processing Systems (NeurIPS), 2020, arXiv:2004.05718.
[41] E. M. Voorhees, "The TREC-8 question answering track report," in Proceedings of the Eighth Text REtrieval Conference (TREC-8), 1999.
[42] B. Efron and R. J. Tibshirani, An Introduction to the Bootstrap. New York: Chapman & Hall, 1993.
[43] F. Wilcoxon, "Individual comparisons by ranking methods," Biometrics Bulletin, vol. 1, no. 6, pp. 80–83, 1945.
[44] S. Bonanno, M. B. Pasanisi, R. Frangiamore, L. Maggi, C. Antozzi, F. Andreetta, A. Campanella, G. Brenna, L. Cottini, and R. Mantegazza, "Amifampridine phosphate in the treatment of muscle-specific kinase myasthenia gravis: a phase IIb, randomized, double-blind, placebo-controlled, double crossover study," SAGE Open Medicine, vol. 6, p. 2050312118819013, 2018.
[45] L. Remijn-Nelissen, W. R. Bakker, T. van Gelder, D. J. A. R. Moes, A. M. Ruiter, Y. J. M. Campman, U. A. Badrising, C. S. M. Straathof, J. J. G. M. Verschuuren, and M. R. Tannemaat, "Efficacy and safety of amifampridine in myasthenia gravis: A randomized, double-blind, placebo-controlled crossover trial," Neurology, vol. 106, no. 8, p. e214715, 2026.